

Name: Jonathan

Group: 99

Project Description

Automated trading is a fascinating field, especially for those who know no better, or the lazy. The concept of leaving your money unattended and having it perform worse than the S&P 500 is truly what puts many minds at ease. This project takes that skepticism seriously. Rather than building yet another price prediction model, this project looks at a simpler question: can knowing what state the market is in actually help a trading strategy do better than just holding the S&P 500?

Markets go through different phases, like bull runs, bear markets, and high volatility periods. The idea is that a strategy which reacts based on what the market is doing should hold up better than one that just stays put. To test this, exploratory data analysis and feature engineering will be applied to raw price data to extract meaningful signals. These features will then serve as inputs to two unsupervised learning methods are compared: Hidden Markov Models and Gaussian Mixture Models. (May add to this if I have more time.)

Scientific Questions

The actual hypothesis comes in two parts. First we assume, Hidden Markov Models should produce more statistically coherent regimes due to its probabilistic sequential nature, unlike its Gaussian Mixture Model counterpart which is purely distribution based. Second, a regime aware trading strategy informed by these detections should outperform a passive S&P 500 hold strategy baseline on raw return and other metrics.

Data Source

<https://docs.alpaca.markets/docs/getting-started>

Alpaca Markets API provides historical and real-time market data and has a generous free tier. We will be sourcing raw market data, covering multiple years of market history to account for exposure. This raw data will serve as the foundation for our feature engineering process, where additional calculations will be made to derive returns, volatility, volume, and price-based metrics for use in our modeling phase.